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PERFORMANCE SPECIFICATION

MODULE, STARTER LOW VOLTAGE PROTECTIVE SYSTEM

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a low voltage protection module to protect engine starter motor.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2920

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods (see 4.2.1).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation

DRAWINGS

ARMY

11668620 - Module, Starter Low Voltage Protective System (interface).

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

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3.2 Design, Materials, and manufacturing processes. Unless otherwise specified, the design, materials, and manufacturing process selection shall be the prerogative of the contractor as long as all articles submitted to the government fully meet the operating, interface, ownership and support , and operating environment requirements specified.

3.3 Operating requirements. Each module shall provide the following functional, operational, and performance capabilities.

3.3.1 Performance. The module shall perform as specified herein through the range of 0-34 volts (V) direct current (dc). Nominal battery voltage to the module shall be 24 V dc. Voltage measurements shall be relative to ground.

3.3.1.1 Battery voltage sensing. The module shall permit continuous variable voltage up to 34 V dc from the battery to the starter. Continuity shall be provided between pins B and D when voltage greater than 11.75 ± 1 V dc is applied to pin A with the module grounded. When less than 11.75 ± 1 V dc from the battery is applied to pin A, continuity between pins B and D shall be interrupted after a 500 ± 50 millisecond (ms) delay. Momentary fluctuations down to below the “break” point, as low as 5 V dc during the 500 ms delay shall not cause continuity between pins B and D to be interrupted. When continuity between pins B and D is interrupted due to low voltage, it shall be re-established and maintained only by removing the power from pin A for 2 ± 1 seconds (s) and applying voltage greater than 11.75 ± 1 V dc again to pin A.

3.3.1.2 Generator voltage sensing. Continuity between pins C and D shall be prevented when voltage applied to pin C is greater than 17 ± 1 V dc. When continuity between pins B and D has been stopped because of high voltage at pin C, the module shall be reset by removing power from pins A and C for 2 ± 1 s and applying voltage greater than 11.75 ± 1 up to 34 V dc to pin A.

3.3.1.3 Power rating. The module shall withstand a continuous current draw of 50 amperes (A) and withstand surges to 100 A (at 24 V dc) through pins B and D.

3.3.1.4 Polarity protection. Performance shall not be impaired by an incident of reverse polarity when the module is operated with a negatively grounded electrical system utilizing electrical storage batteries.

3.3.1.5 Time delay. When 12 V dc or less are applied to pin A with the case grounded, there will be a time delay of 500 ms before interruption of continuity between pins B and D

3.3.1.6 Starter switch bounce. The module shall function normally when a signal that has a simulated switch wave form is applied to pin A.

3.3.1.7 Dielectric strength. Leakage of current through circuit insulation to ground shall not be greater than 100 mA when 1000 volts root mean square (V rms) are applied between the

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case and either pins B or D with the case ground wire disconnected and pins A and C connected to ground.

3.4 Interface requirements.

3.4.1 Envelope, mounting and connection interface. The low voltage protection module shall conform to the envelope dimensions, the bolt hole locations and size and the electrical connection location and interface (see 3.4.4) as specified on drawing 11668620.

3.4.2 Interchangeability. All modules including mounting and connecting provisions shall be completely interchangeable.

3.4.3 Mechanical interface. The module shall be constructed to operate in any mounting attitude.

3.4.4 Electrical interface. Electrical connector pins to the utilizing electrical system shall be designated as follows.

Connector Pin Function	Connector Pin Designation
Battery Voltage Sensor	A
Relay Input	B
Generator Voltage Sensor	C
Relay Output	D
Ground	Case

3.4.5 Weight. The module shall not exceed a weight of 850 grams (g) ((30 ounces)(oz)).

3.5 Ownership and support requirements.

3.5.1 Endurance. The module shall meet the requirements of 3.3.1, and 3.6.1 after a total of 9000 cycles of making at 100 A and breaking at 20 A in a temperature ranging from -54 to 149°C (-65 to 300°F).

3.5.2 Identification and marking. Identification and marking of the module shall be permanent and legible, and shall include, as a minimum.

a. Module, starter low voltage protection.
b. 24 V dc.
c. Identification part number.
d. Manufacturer’s identification number (CAGE).

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3.6 Environmental requirements. The module shall operate under the following environmental conditions, without degradation in performance.

3.6.1 Waterproofness. After being submerged in 5 % of saline solution (5 parts by weight of salt in 95 parts by weight in distilled water).

3.6.2 Shock resistance. After exposure to shock resistance test.

3.6.3 Vibration resistance. During and after exposure to vibrations encountered in the operating environment.

3.6.4 Temperature shock resistance. After exposure to temperature shock.

3.6.5 Salt fog resistance. After exposure to salt fog.

3.6.6 Humidity resistance. After storage under induced high temperature and humidity conditions.

3.6.7 Fungus resistance. After exposure to fungal growth.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1).
b. Conformance inspections (see 4.3.2).

4.2 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives that replace verifications required by the specifications.

4.3 Order of inspection. The inspection sequence may be in any order.

4.3.1 First article inspection. First article inspection shall be performed on the first four production samples as specified in table I.

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TABLE I. Order of first article test.

Sample 1	
Battery voltage sensing	4.5.1
Generator voltage sensing	4.5.2
Polarity protection	4.5.4
Time delay	4.5.5
Starter switch bounce	4.5.6
Waterproofness	4.8.1
Shock resistance	4.8.2
Salt fog resistance	4.8.5
Fungus resistance	4.8.6
Battery voltage sensing	4.5.1
Generator voltage sensing	4.5.2
Time delay	4.5.5
Sample 2	
Battery voltage sensing	4.5.1
Generator voltage sensing	4.5.2
Polarity protection	4.5.4
Time delay	4.5.5
Waterproofness	4.8.1
Vibration resistance	4.8.3
Battery voltage sensing	4.5.1
Generator voltage sensing	4.5.1
Time delay	4.5.5
Sample 3	
Battery voltage sensing	4.5.1
Generator voltage sensing	4.5.2
Polarity protection	4.5.4
Time delay	4.5.5
Waterproofness	4.8.1
Shock resistance	4.8.2
Humidity resistance	4.8.6
Battery voltage sensing	4.5.1
Generator voltage sensing	4.5.2
Time delay	4.5.5

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TABLE I. Order of first article test - Continued.

Sample 4	
Battery voltage sensing	4.5.1
Generator voltage sensing	4.5.2
Power rating	4.5.3
Polarity protection	4.5.4
Time delay	4.5.5
Waterproofness	4.8.1
Temperature shock	4.8.4
Endurance	4.7.1

TABLE II. Verification methods.

Title	Requirement	Verification
Operating requirements	3.3	4.5
Battery voltage sensing	3.3.1.1	4.5.1
Generator voltage sensing	3.3.1.2	4.5.2
Power rating	3.3.1.3	4.5.3
Polarity protection	3.3.1.4	4.5.4
Time delay	3.3.1.5	4.5.5
Starter switch bounce	3.3.1.6	4.5.6
Dielectric strength	3.3.1.7	4.5.7
Interface requirements	3.4	4.6
Envelope, mounting and connection interface	3.4.1 thru 3.4.5	4.2
Ownership and support requirements	3.5	4.7
Endurance	3.5.1	4.7.1
Identification and marking	3.5.2	4.7.2
Environmental requirements	3.6	4.8
Waterproofness	3.6.1	4.8.1
Shock resistance	3.6.2	4.8.2
Vibration resistance	3.6.3	4.8.3
Temperature shock resistance	3.6.4	4.8.4
Salt fog resistance	3.6.5	4.8.5
Humidity resistance	3.6.6	4.8.6
Fungus resistance	3.6.7	4.8.7

1/ SIE = Standard Inspection Equipment.

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4.3.2 Conformance inspection. Conformance inspection shall include the examinations and tests from table II as defined in the contract.

4.4 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 22.8 ± 10°C (73 ±18°F)

b. Barometric pressure: 96.5 (+6.8, -15.2) kilopascal (kPa) (28.5 (+2, - 4.5) inches (in)) Hg of mercury

c. Relative humidity: 50 ±30 percent.

4.5 Operating requirements verification.

4.5.1 Battery voltage sensing. The module shall be connected to 0-34 V dc variable voltage power source with voltmeter and lamp as shown in figure 1. Connect pins A and B to source, pin D to lamp, while pin C is unconnected. With power supply set at 14 V dc or above, the switch shall be closed to energize unit. Wait at least 500 ms to allow “ON” delay timer to time out. Raise voltage to 34 V dc observing lamp action. Lower voltage slowly until lamp goes out. Record voltage at which lamp goes out. Continue to reduce to 4 V dc. Increase voltage to 34 V dc, lamp should not relight.

4.5.2 Generator voltage sensing. The module shall be connected to 0-34 V dc variable voltage power source with voltmeter and a lamp as shown in figure 1. Connect pins A, B, and C to source, and pin D to lamp (Note: this differs from figure 1 - reference 4.2). With power supply set to 15 ± 1 V dc close switch energizing unit. No time delay required. Raise supply voltage slowly until lamp turns off. Record voltage. Raise voltage to 34 V dc observing lamp action. Reduce voltage slowly to 14 V dc.

4.5.3 Power rating. The module shall be tested in a circuit as shown in figure 2 for 3000 cycles at a rate of 17 minutes per cycle. The current draw shall be 50 A for duration of 15 min followed by a surge to 100 A for a duration of 70 ms.

4.5.4 Polarity protection. The module shall be connected with positive potential connected to the case and negative potential connected to pins A, B and C for one minute Reconnect pin as specified in 3.4.4. Verify performance of 3.3.1.1 and 3.3.1.2 to assure conformance to 3.3.1.4.

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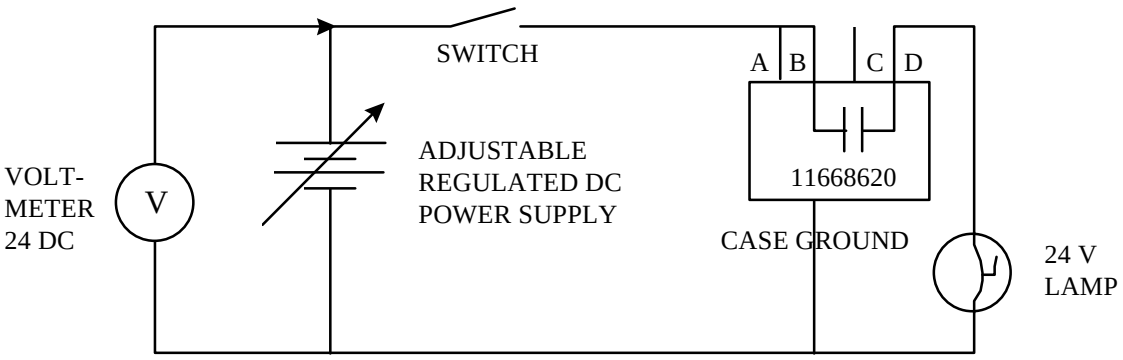


FIGURE 1. Schematic for voltage tests.

4.5.5 Time delay. The module shall be connected to 0-34 V dc variable voltage power source per figure 3 with voltmeter and oscilloscope (or electronic counter). Preset power supply voltage to 11.25 ± 0.25 V dc. Close switch, energizing module and initiating scope trace sweep. When internal relay drops out, scope trace indicates time delay. See figure 3.

4.5.6 Starter switch bounce (figure 3). Apply 24 V dc with an initial simulated switch bounce wave form of 6 ms duration to pin A of the module with the case grounded. Continuity shall exist between pins B and D.

4.5.7 Dielectric strength . The assembly shall be tested as follows:

a. Connect pins A and C to the case (ground) (reference 3.4.4).
b. Apply 1000 V rms from pin B to case.
c. Apply 1000 V rms from pin D to case.

Verify that leakage of current through circuit insulation to ground is not greater than 100 milliamperes (mA).

4.6 Interface requirements verification. Interface requirements shall be verified using one or more of the methods specified in 4.2.

4.7 Ownership and support verifications. Ownership and support verifications, except as specified below, shall be verified using one or more of the methods specified in 4.2.

4.7.1 Endurance. After exposure to waterproofness test of 4.8.1, when connected into a circuit as shown on figure 2, and 24 V dc are applied to pin A, the module shall be operated for 3000 cycles making at 100 A and breaking at 20 A at each of three ambient: -54°C (-65°F) +24°C (+75°F), and +149°C (total of 9000 cycles). Verify that the module conforms to 3.5.1.

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4.7.1.1 Endurance control. When connected in a circuit using the battery configuration as shown in figure 7, operate the module per table III for 400 cycles. Then connect the circuit, using the generator configuration as shown in figure 7, operate the circuit per table IV for an additional 400 cycles. Then, verify conformance to 3.5.1 by testing as specified in 4.8.1.

4.7.2 Identification and marking. Inspect module for improper or illegible marking.

4.8 Environmental conditions verifications. Module shall meet the operating performance requirements of 3.3 after exposure to the conditions of 4.8.1 through 4.8.7.

4.8.1 Waterproofness. The module shall be submerged in a container with the uppermost surface a minimum of one inch below the surface of 5% of saline solution, and installed in the chamber. The module shall be observed during its entire period of submersion and shall be operated while submerged for 30 minutes at full rated current and voltage. the chamber shall be evacuated to a pressure 68.94 kPa (10 pounds per square inch (psi)) below atmospheric so as to apply a minimum of 68.94 kPa (10 psi) internal pressure to all voids within the module. During this period the module shall be carefully observed for poor seals, as evidenced by bubbles escaping from the interior of the module. Leakage thus indicated shall be considered as noncompliance with the waterproofness requirement and the module shall be rejected. Bubbles which are the results of entrapped air on the exterior surfaces of the components shall not be considered a leak. Repeat the same process while the module pressurized to ten lbs above atmospheric.

4.8.1.1 Waterproofness for acceptance. The module shall be exposed to the test and verification of 4.8.1 except that tap water shall be used in lieu of salt water.

4.8.2 Shock resistance. The module shall be mounted to the test fixture in the installed interface configuration and subjected to a saw tooth pulse of 20 ± 2 gravities for 11 ± 1.1 ms three times in each of the three orthogonal axes for a total of 18 shock pulses. The saw tooth pulse configuration and tolerance limits shall be as shown for ground equipment in figure 516.4-4 of MIL-STD-810, or equivalent (see 4.2.1).

4.8.3 Vibration resistance. Modules shall be solid mounted to the vibration fixture in the installed interface configuration using the bolt holes of the mounting flange. Sinusoidal cycling time shall be 30 min/1609 kilometers (km) (1000 miles). Maximum cycling shall be 3 hours (hr). Each sweep shall be 5 to 500 to 5 hertz (Hz) in 15 min. The test shall be conducted in each of the three orthogonal axes. Amplitude and test level shall be in accordance with the curve of figure 4. Resonant frequencies shall be determined and a sinusoidal resonance search shall be conducted for the four most severe resonant frequencies of the module. The module shall be subjected to each severe resonant frequency within each 60 min sweep test. The frequency of applied vibration shall be swept over the specified range in accordance with figure 5. The sweep time is that of an ascending plus a descending time shown in figure 5.

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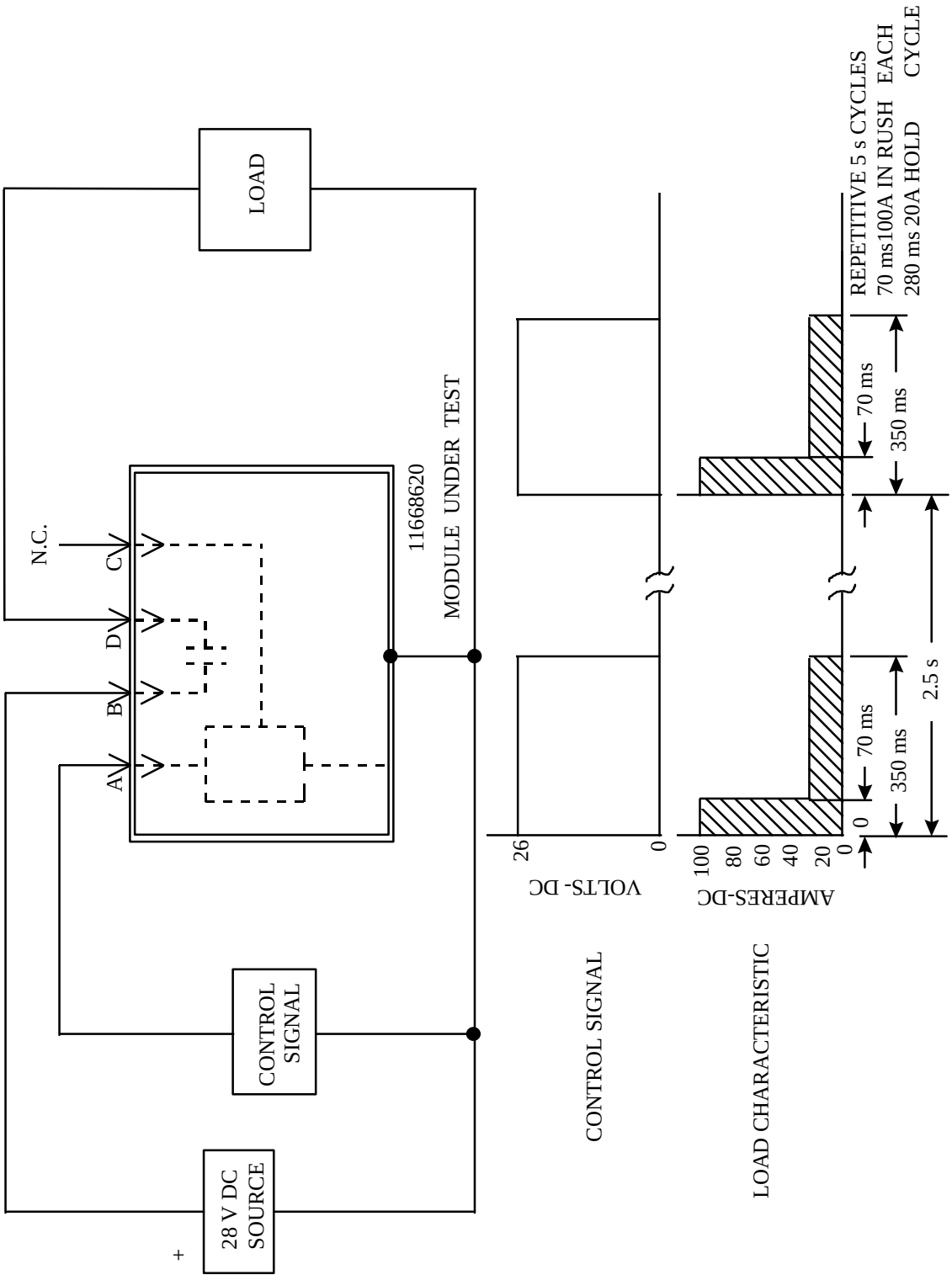


FIGURE 2. Endurance (level B).

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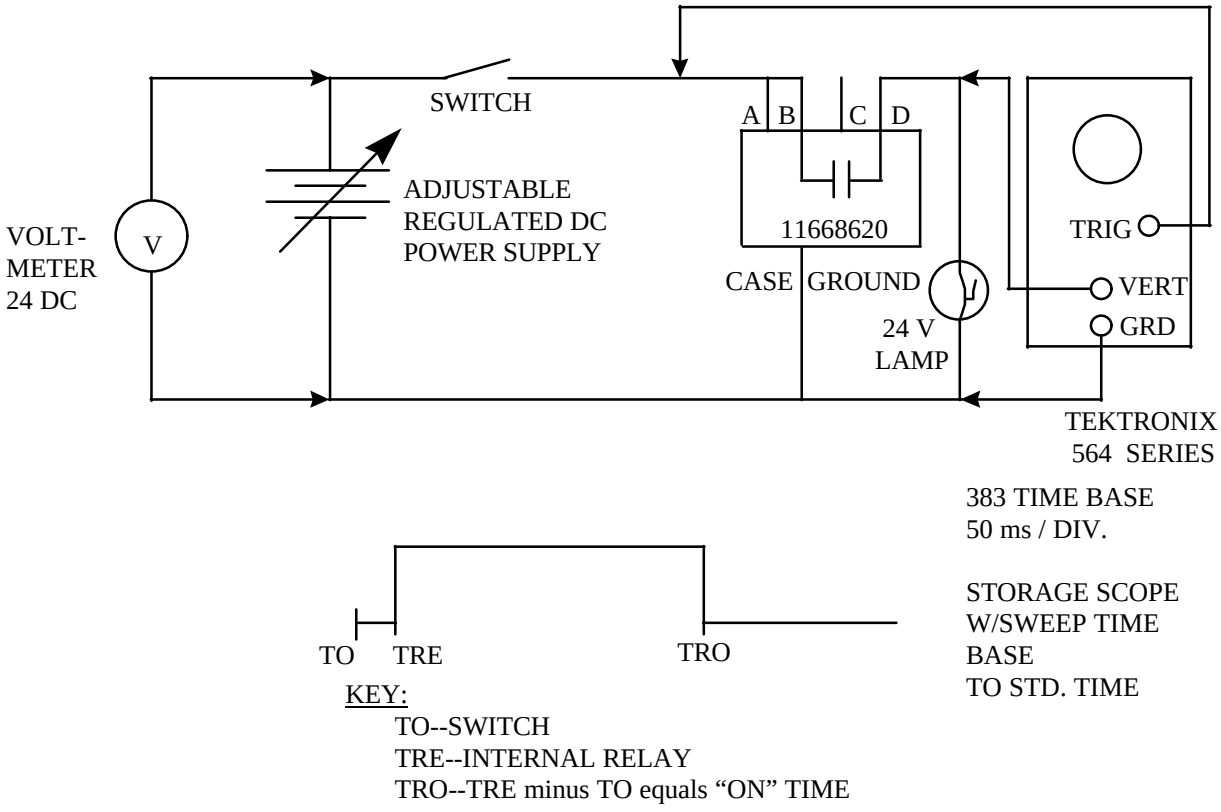


FIGURE 3. Schematic for time delay test.

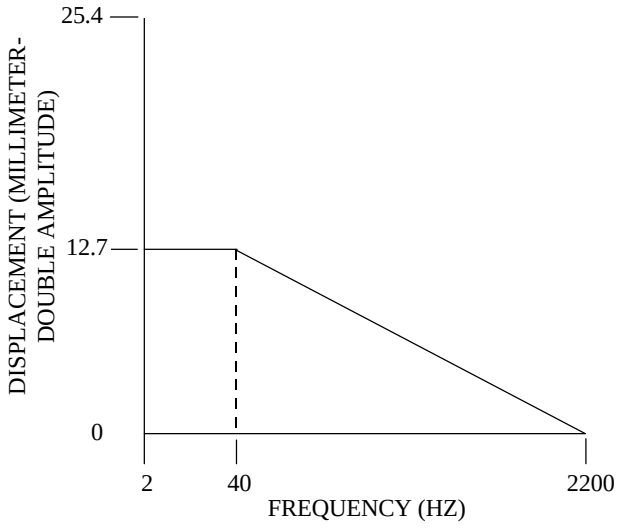


FIGURE 4. Vibration test curve.

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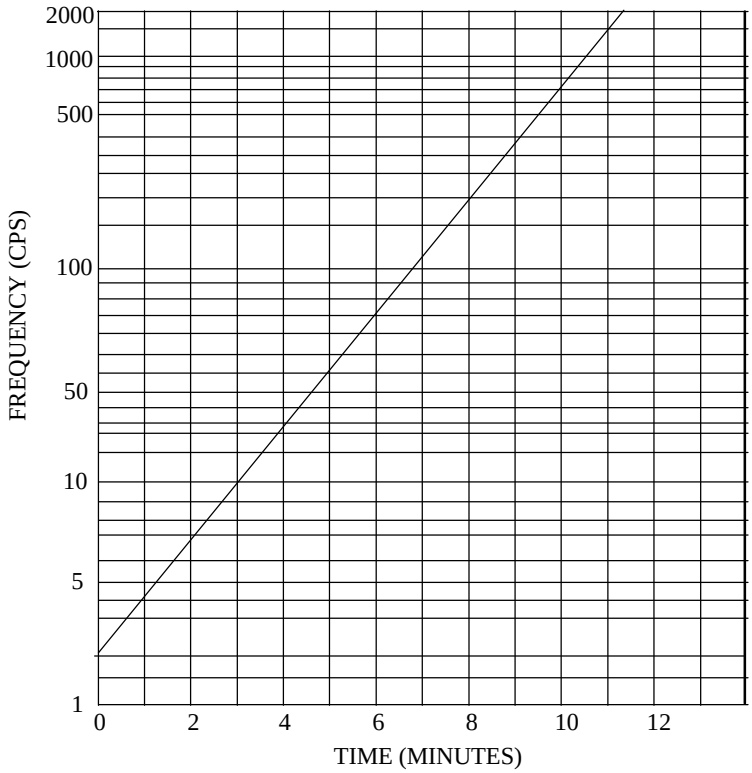


FIGURE 5. Logarithmic sweep.

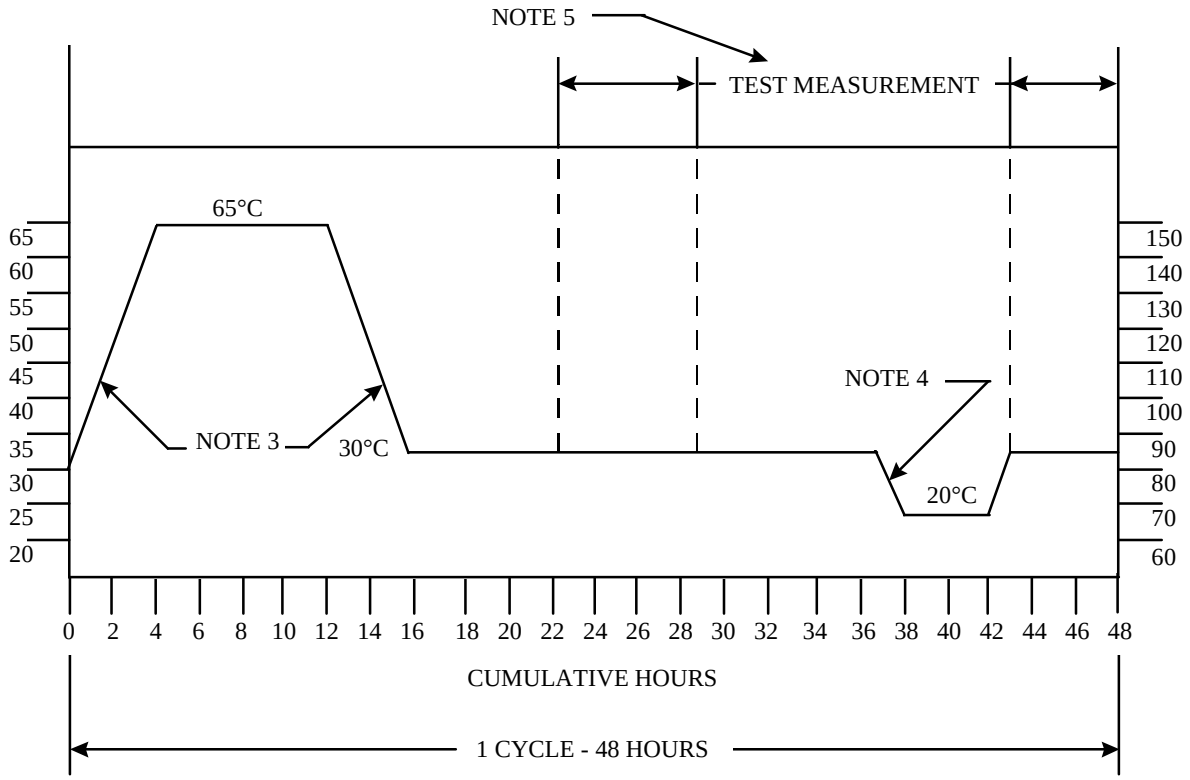
4.8.4 Temperature shock resistance. The module shall be subjected to a temperature shock, without any electrical energy applied, in accordance with method 503.3, section II of MIL-STD- 810, or equivalent (see 4.2.1). The internal temperature of the cold chamber shall be -57°C (-70°F). The internal temperature of the high temperature chamber shall be +149°C The module shall show no degradation of performance when tested in accordance with 4.5.1, 4.5.2, 4.5.5, 4.5.6, 4.5.7, and 4.8.1.

4.8.5 Salt fog resistance. The module shall be sealed so that no internal components are exposed and subjected to salt fog as specified in method 509.3, section II of MIL-STD-810, or equivalent (see 4.2.1), except the exposure to salt fog shall be for 192 hr. The module shall perform as specified herein when tested in accordance with 4.5.1, 4.5.2, 4.5.5, 4.5.7, and 4.8.1.

4.8.6 Humidity resistance. The module shall be exposed to warm high relative humidity in accordance with applicable procedure for ground electronic equipment of method 507.3 of MIL-STD-810, or equivalent (see 4.2.1). After conditioning, the module shall be exposed to five continuous 48 hr cycles in accordance with figure 6. Prior to operation, after exposure, the module shall be conditioned for 24 hr at 23°C and 50 ± 10 percent relative humidity.

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4.8.7 Fungus resistance. The module shall be exposed to 28 days fungus incubation in accordance with method 508.4 of MIL-STD-810, or equivalent (see 4.2.1). The module shall evidence no operational degradation when tested in accordance with 4.5.1, 4.5.2, 4.5.5, 4.5.6, 4.5.7, and 4.8.1.

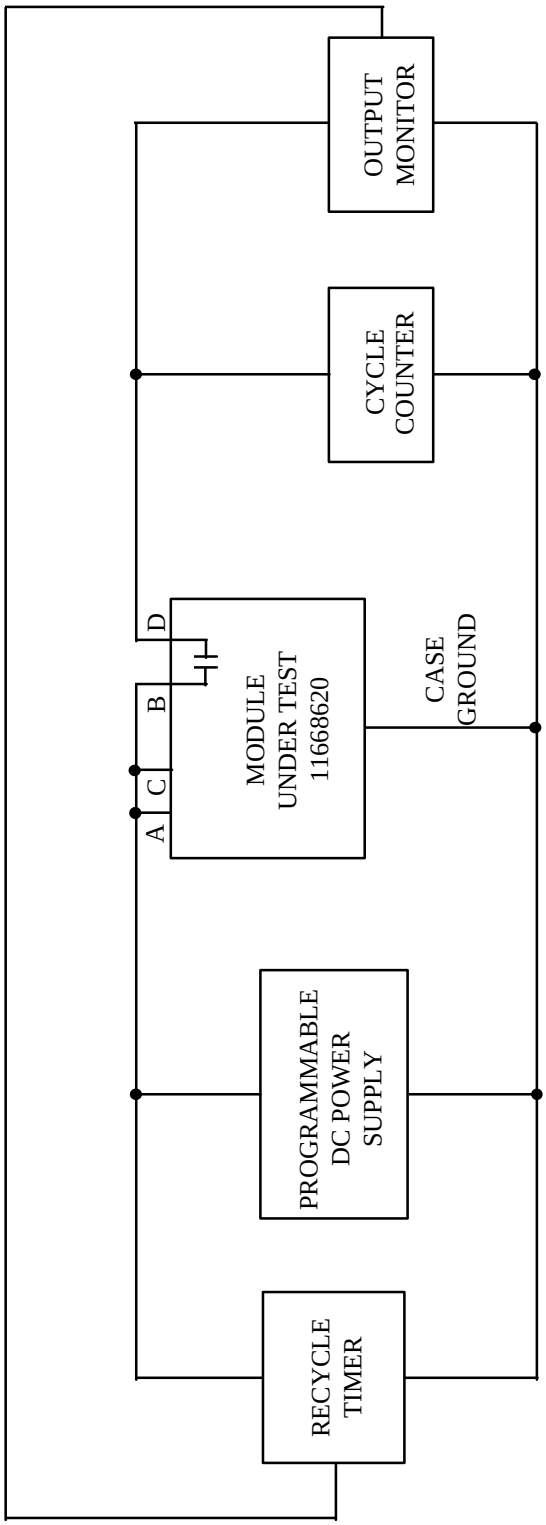


NOTES:

1. Tolerance during temperature change shall not be greater than 3°C (5°F).
2. Relative humidity shall be maintained at 94 ± 4 percent at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85 percent.
3. Rate of temperature change between 30°C and 65°C (86°F and 149°F) shall not be less than 8°C (14.4°F) per hour.
4. The temperature increase in this portion of the curve shall not be less than 10°C (18°F).
5. Test measurement shall be taken only at the period specified in the applicable equipment or system specification.

FIGURE 6. Humidity cycle.

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RECYCLE TIMER PROGRAMS POWER SUPPLY OUTPUT FOR 15 ± 1, 20 ± 1, AND 0 VDC AS TIMER CYCLES.

11668620 ENDURANCE TEST APPLIED TO GENERATOR / BATTERY VOLTAGE SENSE FUNCTION.

MODULE UNDER TEST IS ENERGIZED AT 15 VDC FOR 4 ± 1 SEC. IT MUST REMAIN CLOSED DURING THIS TIME. THE APPLIED VOLTAGE IS THEN INCREASED TO 20 VDC FOR 4 ± 1 SEC. IT MUST DROP OUT IMMEDIATELY UPON APPLICATION OF 20 VDC AND REMAIN OPEN UNTIL NEXT CYCLE. THE VOLTAGE IS REDUCED TO 0 VDC FOR 2 ± 1 SEC. FOR RESET TO COMPLETE THE CYCLE. THE OUTPUT MONITOR WILL STOP THE TEST IF ABOVE CONDITIONS FAIL TO BE MET.

FIGURE 7. Test, generator / battery voltage sensing

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TABLE III. Endurance cycles - voltage (battery).

Applied dc voltage, pin A+	Time (seconds)	Continuity between pins B and D
14 ± 1 V dc	5	Yes
10 ± 1 V dc	5	NO (A nominal 500 ms closure may occur)
	Total Endurance = 400 cycles of 10 s each.	

NOTES: 1. C open
2. Case is negative and grounded.

TABLE IV. Endurance cycles - voltage (generator).

Applied dc voltage, at pins A,B, and C+	Time (seconds)	Continuity between pins B and D
15 ± 1 V dc	5	Yes
20 ± 1 V dc	5	NO (A nominal 500 ms closure may occur)
	Total Endurance = 400 cycles of 10 s each.	

NOTE: Case is negative.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The Module, Low Voltage Protective System covered by this specification is intended to protect electrical starters for AVDS 1790 series engines from damage due to low battery voltage. This System is military unique because it must be able to operate satisfactorily under steady accelerations of 20 g applied in any direction; at an ambient temperatures ranging from -70°F to +300°F. Commercial Modules are not designed to withstand such extreme and sudden environmental conditions and would experience catastrophic failure.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
c. If first articles samples are not required (see 3.1).
d. Applicable drawing (see 3.3)
e. If inspection conditions should be other than as specified (see 4.2).
f. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Battery
Generator
Sensor
Relay

6.4 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a

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first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitations documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.5 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitations, contracting documents should indicate those tests desired from Table II and their designated frequency based on a risk assessment for the procurement.

6.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

Custodian:
Army - AT

Preparing activity:
Army - AT

Review activity:
DLA - CC

(Project 2920-0483)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-62539A	**2. DOCUMENT DATE *(YYMMDD)*** 971231

3. DOCUMENT TITLE

Module, Starter Low Voltage Protective System

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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